

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/264350212>

A review of Beijing's vehicle registration lottery: Short-term effects on vehicle growth and fuel consumption

Article in *Energy Policy* · July 2014

DOI: 10.1016/j.enpol.2014.05.055

CITATIONS

25

READS

1,829

4 authors:



Jun Yang

Peking University

7 PUBLICATIONS 53 CITATIONS

[SEE PROFILE](#)



Ying Liu

Beijing University of Technology

1 PUBLICATION 25 CITATIONS

[SEE PROFILE](#)



Ping Qin

Renmin University of China

49 PUBLICATIONS 653 CITATIONS

[SEE PROFILE](#)



Antung Liu

Cheung Kong Graduate School of Business

8 PUBLICATIONS 36 CITATIONS

[SEE PROFILE](#)

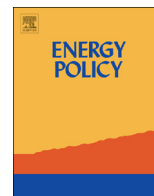
Some of the authors of this publication are also working on these related projects:



Chinese National Household Energy Survey [View project](#)



The Effects of Urban Rail Transit on Air Quality: New Evidence from Multiple Chinese Cities" [View project](#)



A review of Beijing's vehicle registration lottery: Short-term effects on vehicle growth and fuel consumption

Jun Yang^a, Ying Liu^b, Ping Qin^c, Antung A. Liu^{d,*}

^a College of Environmental Science & Engineering, Peking University, Beijing, China

^b College of Urban Transportation, Beijing University of Technology, Beijing, China

^c Department of Energy Economics, School of Economics, Renmin University of China, Beijing, China

^d Resources for the Future, Washington, DC, USA, and Cheung Kong Graduate School of Business, Beijing, China

HIGHLIGHTS

- In 2011, Beijing became the first city to allocate vehicle license plates using a lottery.
- We provide a background on Beijing's lottery and analyze its short-term effects.
- Beijing's lottery has sharply curtailed new vehicle registrations and reduced the growth of cars in Beijing.
- While the number of vehicles will be cut 11 percent, fuel consumption will be reduced by only 1 percent.

ARTICLE INFO

Article history:

Received 24 March 2014

Received in revised form

23 May 2014

Accepted 30 May 2014

Keywords:

China energy

Beijing lottery

transportation policy

fuel consumption policy

vehicle restrictions

ABSTRACT

Many cities worldwide have considered vehicle restriction policies to curb proliferating problems related to traffic and pollution. At the beginning of 2011, Beijing became the first city to allocate vehicle license plates using a lottery. We provide a background on Beijing's lottery and analyze its short-term effects. We find that growth in new vehicle registrations has been sharply curtailed. However, this policy may not reduce fuel consumption as much as expected.

© 2014 Elsevier Ltd. All rights reserved.

1. Introduction

China's economic development has created a new set of problems related to the rapidly growing number of vehicles. Although the wider availability of cars has improved the standard of living of China's residents, it has also exacerbated problems of air pollution, resulted in intense traffic congestion, and driven rapid growth in the demand for energy.

In January 2011, Beijing began restricting the purchase of automobiles by distributing new license plates via a publicly held lottery. Although the concept of restricting license plates is no longer new, Beijing is unique in its use of a lottery for this purpose rather than an auction system.

Singapore was the first country to implement license plate restrictions. Singaporeans participate in a monthly auction for the right to purchase a car. Revenue from these auctions is used to improve roads and public transportation. Following Singapore's example, Shanghai implemented a similar auction system in 1994. [Chen and Zhao \(2013\)](#) surveyed Shanghai residents and concluded that they held negative attitudes toward the fairness of the auctioning policy.

As of January 2014, four Chinese cities have implemented license plate restrictions: Beijing, Shanghai, Guangzhou, and Guiyang. Guangzhou began restricting car purchases using a system with hybrid auction and lottery components. Many more cities in China are also considering restrictions. Internationally, many cities struggling with traffic are considering various models of restricting the use of automobiles.

We provide an in-depth look at Beijing's experience using a license plate lottery to allocate vehicles. We highlight in this paper three key differences between using a lottery rather than an auction to allocate cars when the supply has been restricted.

* Corresponding author. Tel.: +86 10 8518 8858.

E-mail addresses: yangj@bjtrc.org.cn (J. Yang), liuying@bjjtw.gov.cn (Y. Liu), pingqin2009@gmail.com (P. Qin), liu@rff.org, antung@ckgsb.edu.cn (A.A. Liu).

<http://dx.doi.org/10.1016/j.enpol.2014.05.055>

0301-4215/© 2014 Elsevier Ltd. All rights reserved.

First, an auction raises revenue, while a lottery does not. [Chen and Zhao \(2013\)](#) document that 5 billion renminbi (RMB) were raised through the Shanghai auction in 2011; Beijing's lottery does not raise revenue. In Shanghai, this revenue was allocated to developing roads and building out public transportation. [Li \(2014\)](#) estimated that the auction could have brought in 21 billion RMB in revenue in 2012, greater than the cost of all subsidies to the public transit system.

Second, in Beijing, many license plate lottery winners did not use their "quotas" (permits allowing for a vehicle purchase) and instead allowed their quotas to expire without buying a car; in some months, more than 10 percent of lottery winners allowed their quotas to expire. It is difficult to imagine auction entrants purchasing expensive permits and then allowing them to expire. As a result, vehicles grow somewhat more slowly than is permitted under the quota system.

Third, in an auction system, those with the highest willingness to pay receive cars. In a lottery system, by contrast, the recipients of vehicles are a random subset of entrants. Therefore, the lottery system will allocate vehicles to people who do not have the highest willingness-to-pay. [Li \(2014\)](#) estimated just the costs from this mis-allocation, and found that the use of the lottery policy cost Beijing 36 billion RMB in social welfare in 2012. Another consequence of allocating license plates to those who do not have the highest willingness to pay is a rise in the black market for license plates. We discuss below two examples where individuals could illegally pay large sums of money to rent license plates from other individuals or from car dealerships.

The primary purposes of this paper are to present a background on Beijing's lottery policy, and to provide an analysis of its short-term impacts. We first provide a brief history of Beijing's vehicle regulation efforts. We then discuss how Beijing's lottery is implemented, and the intended and unintended consequences of its implementation.

Our paper is the first to study the impact of the license plate lottery on trends in automobile use in Beijing. We add to an extensive literature studying China's car market and projected demand for fuel. Because of China's size and growth rates, its automobile market is currently the largest and most important in the world, with an outsized impact on global energy consumption patterns. In the most directly comparable paper, [Hao et al. \(2011\)](#) compared policies in Beijing and Shanghai to study their impact on fuel consumption. This paper was written before Beijing's lottery had been planned, and does not attempt to evaluate the effects of the lottery.

We find that, in the first few years after adoption of the policy, growth in the number of cars in Beijing has been sharply reduced. When combined with other traffic-easing policies, congestion in Beijing dropped at the same time this policy was implemented. Finally, we use a few simple calculations to provide rough estimates of the lottery's impact on fuel consumption. We consider two effects which partially offset the impact of reducing the number of cars on fuel consumption. First, cars are driven more intensively when congestion improves. Second, buyers in a restricted-license plate regime tend to concentrate their purchases on larger, less fuel efficient cars. As a result, our central result finds that although the lottery will reduce the number of vehicles by 11 percent from a no-policy scenario by 2020, this policy will reduce fuel consumption by only 1 percent.

2. Why did Beijing implement a lottery?

2.1. The need for restrictions on car use

China's rapid economic development is reflected in the rise in incomes of the residents of its capital Beijing. Beijing's gross

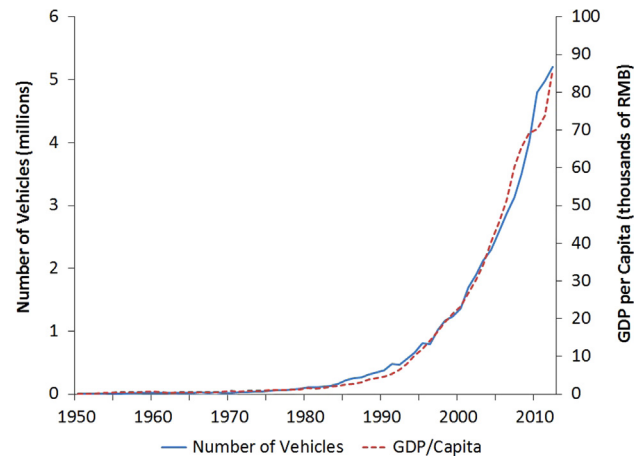


Fig. 1. Growth in the number of vehicles in Beijing and in GDP per capita. Source: Beijing Municipal Commission of Transport and Beijing Transportation Research Center (2012).

domestic product (GDP) per capita has shot up in recent decades. In 2000, GDP per capita was US\$2915. In 2010, it was US\$11,218, an annual growth rate of 14 percent ([Fig. 1](#)).

Along with its growth in GDP per capita, Beijing's population has grown rapidly. Between 1990 and 2000, its population grew 26 percent, and between 2000 and 2010, its population grew a further 44 percent. In 2010, its population of permanent residents reached 19.6 million.

The municipality of Beijing officially occupies an area of 16,400 square kilometers (km), roughly the size of Connecticut. Most of Beijing's population and vehicles are concentrated in the central city area enclosed by the concentric rings forming the city's primary highway system. The rapidly growing population and limited prime real estate have led to a densely packed downtown. Land area per capita within Beijing's core commercial area, enclosed by the 2nd Ring highway, is only 44 m² per person.¹ About 84 percent of vehicles are distributed within the 6th Ring highway.

The rapid growth in population and per capita wealth has been reflected in vehicle purchases. Beijing took seven years to advance from 1 million cars to 2 million cars, but only four more years to reach 3 million, and three more years to reach 4 million. In 2005, 2.6 million cars were registered in Beijing; by 2010, the number of vehicles had reached 4.8 million, an annual growth rate of 13 percent. Between 1950 and 2010, the growth in GDP per capita was closely mirrored by the growth in the number of cars ([Fig. 1](#)).

The rapid increase in cars was accompanied with the rapid expansion of existing roads and the construction of new road networks, yet the number of new vehicles began to overwhelm the city's best efforts. Beijing has been ranked by numerous international organizations as one of the most congested cities in the world (see, for example, [IBM, 2010](#)). The average speeds of vehicles on arterial roads during the morning rush hour decreased from 30 km/h in 2003 to 20 km/h in 2010, with even larger drops in vehicle speeds on secondary roads ([Beijing Transportation Research Center, 2002–2013](#)).

Beijing's automobile congestion has large and growing costs. Using data on vehicle flow rates and speeds, [Xie et al. \(2011\)](#) estimated that the costs of time delay, fuel consumption, and air pollution were between RMB 5 and 25 billion in 2008, about 0.5 to 2.5 percent of Beijing's GDP.

¹ By comparison, the central city portion of the New York metropolitan area has a land area per capita of 97 m² per person ([Manville and Shoup 2005, Table 3](#)).

Moreover, Beijing's growing fleet of cars emits massive amounts of carbon monoxide, nitrogen oxide, and volatile organic compounds, contributing to Beijing's painfully poor air quality. According to data from the Beijing Municipal Environmental Protection Bureau, motor vehicle exhaust accounts for 40–60 percent of suspended particulates, 80 percent of hydrocarbons, 70–80 percent of nitrogen oxide, and 23 percent of coarse particulate matter (PM₁₀). Liu et al. (2008) estimated that motor vehicles are the second largest source of PM_{2.5} in Beijing.

2.2. Other attempts to control congestion

Beijing's government has adopted a series of administrative policies to relieve the stresses caused by congestion. These policies have taken a number of different forms; we outline them in rough chronological order here.

As early as the 1990s, Beijing began taking measures to restrict the total number of motor vehicles on city roads. Beginning on March 1, 1998, Beijing imposed a parking certificate system on newly purchased motor vehicles in eight urban districts. This system required all motor vehicle owners applying for a license plate in the urban and inner suburban districts of Beijing to submit parking certificates checked by local public security and transportation administration authorities. This system was designed to restrict the number of vehicles in urban districts and to relieve congestion. However, the system was largely ineffective because of the slow registration and administration of parking certificates, the high costs of verification, and the arbitrary collection of fees. In addition, some parking certificates were forged. On January 1, 2004, the parking certificate system was officially canceled.

In early 2004, the parking restriction system was replaced by a vehicle purchase tax, promulgated by the Interim Regulations of People's Republic of China on Vehicle Purchase Taxes (The Central People's Government of the People's Republic of China, 2001).² Under these regulations, a tax is levied on all organizations and individuals within the boundaries of China who purchased designated vehicles, including motor vehicles, motorcycles, tramcars, trailers, and agricultural trucks. Despite a tax of 10 percent on the purchase of cars, Beijing continued to experience high rates of growth in motor vehicle purchases.

Starting in 2005, Beijing sharply increased its investments in fundamental transportation facilities. In 2010, RMB 72 billion were invested, over 5 percent of Beijing's GDP. Of this amount, RMB 7.7 billion were invested in the subway system, RMB 9.7 billion in roads, and RMB 41.1 billion in urban public transportation. Beijing's subway system has expanded rapidly. At the end of 2009, subway tracks stretched 228 km. In 2010, 2011, and 2012, the distance covered was increased by 108 km, 36 km, and 70 km, respectively.

In 2007, Beijing implemented a low-cost public transportation policy, which reduced bus fares by 60 percent, with large reductions in subway fares as well. This policy was intended to increase the attractiveness of buses and the subway system. As a result, the fares of most buses and the subway decreased to just RMB 0.4 and RMB 2, respectively. However, this policy does not appear to have had a substantial impact on travel habits. A survey conducted by the website People's Daily Online found that very few Beijing citizens actually changed their typical modes of transportation despite the lower fares.

Because these market-based systems appeared to have only limited impact, Beijing has more recently relied on administrative means to restrict the growth in vehicles. During the Olympics in August 2008, Beijing implemented a license plate restriction system

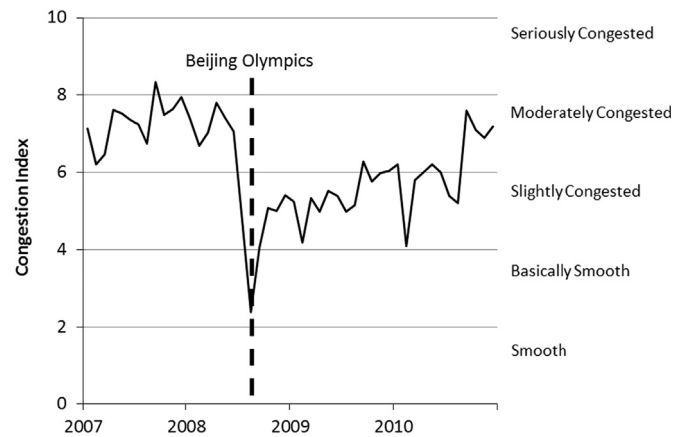


Fig. 2. Congestion index for Beijing (2007–2010).

Source: Beijing Municipal Commission of Transport and Beijing Transportation Research Center (2012).

in which cars could be driven on the road only on certain days according to whether they had an even- or odd-numbered plate.

Before the Olympics, the monthly congestion index consistently averaged above 7, reflecting “moderate” congestion.³ The new restrictions had an immediate effect, sharply reducing congestion (Fig. 2) and, in combination with other simultaneous reforms, achieving large reductions in air pollution.

Following the Olympics, Beijing implemented an ongoing license plate restriction system in which each car was barred from Beijing roads for one weekday each week, depending on the last digit of the license plate number. Congestion remained improved in the early months of 2008, and measured levels averaged below 6 for most months of 2009. However, the rapid increase in vehicle registrations in Beijing during 2009 and 2010 gradually worsened traffic until, by the end of 2010, the congestion index had returned to the level observed prior to implementation of the license plate restriction.

In some ways, Beijing's experience with license plate restrictions mirrored that of Mexico City, which in 1989 introduced a similar program banning vehicles one workday each week. Many citizens in Mexico City circumvented the policy by purchasing an additional car, causing second-hand vehicle sales to soar in the market. Until 1995, 22 percent of drivers bought second vehicles (Eskeland and Feyzioglu, 1997), resulting in increased traffic congestion. Davis (2008) found that the policy restriction did not improve air quality in Mexico City; instead, air pollution worsened during weekends and at off-peak periods during weekdays.

Similarly, in Beijing, drivers gradually found ways to circumvent the license plate restrictions. Drivers took detours and avoided restricted periods. In some cases, drivers purchased multiple cars with different license plate numbers to avoid restrictions on their driving. Each of these measures eroded the usefulness of the license plate restrictions.

In summary, prior to the lottery, each of Beijing's attempts to control its congestion was either ineffective or only temporarily effective. Market-based reforms, such as increasing the price of parking or taxing the sales of cars, were inadequate to dent the enormous demand generated by Beijing's growth. Even restrictions on driving days seemed to only delay the overwhelming problems of dense traffic.

³ The Beijing Municipal Commission of Transport classifies traffic congestion according to a 10-point, 5-level congestion index (Fig. 3). According to this index, congestion is qualitatively described as “smooth,” “basically smooth,” “slightly congested,” “moderately congested,” and “seriously congested.”

² Deng and Ma (2010) estimated an own-price elasticity of −9.2 for autos in China using annual data from 1995 to 2001.

3. Implementation of the Beijing lottery

Beijing's decision to impose restrictions on the number of cars sold each year was perhaps precipitated by a pair of embarrassing incidents. On September 14, 2010, construction work on roads between Beijing and neighboring Zhangjiakou triggered a paralyzing traffic jam stretching over 100 km on the highway just outside of Beijing. For three days, traffic came to a standstill, with those affected progressing as little as half a kilometer per day. Then, on September 17, 2010, a traffic jam triggered by heavy rain affected the entire city. The congestion index remained at 10 for three consecutive hours, and 4321 roads were seriously congested during the evening peak period. The jam lasted until about 1:00 a.m. the next day.

No announcement or report documents the official reasons why Beijing chose a lottery, rather than an auction, to restrict vehicle growth. For years before Beijing's policy, both Singapore and Shanghai had successfully restricted cars with auctions. However, based on our interviews with administrators within Beijing's transportation agencies, we suggest two reasons.

First, a lottery system is seen as more equitable. A resident of average income would have a greater chance of obtaining a license plate through a lottery than he might if plates were sold through an auction. Income equality has become an increasingly important issue in China, particularly to its urban residents. Car ownership is not seen as a luxury for the rich, but rather a symbol of entering the middle class. Second, Beijing had used a lottery in the past to allocate affordable housing. This policy, which was seen as successful, may have inspired the use of a lottery to allocate vehicles.

On December 13, 2010, Beijing first issued a plan to its residents that hinted at the upcoming cap to vehicle registrations. The plan also included higher parking fees, expanded public transportation options, and the construction of new roads. Starting on December 24, 2010, new registrations were frozen, and residents were informed that, to purchase a license plate, individuals and corporations would be required to win a publicly held lottery administered by the Beijing Municipal Commission of Transport.

The lottery is held on the 26th day of each month. Residents compete for 17,600 license plates each month. To be entered into the lottery, applicants register on a government website with no monetary cost. The application consists of 10 questions, including the applicant's registration number, address, and contact information. Applicants are screened to make sure they are qualified to enter the lottery. Only those with a Beijing registration or residents who have paid taxes in Beijing for five years are allowed to enter the lottery. In addition, individuals who already have vehicles may not enter the lottery. However, a second driver in a household with a car may enter the lottery to win the right to buy a second car for the household.

Businesses can also win vehicle license plates, entering a separate lottery with 2400 winners per month; 400 of these license plates are reserved for businesses that rely on fleets of commercial cars, such as rental car and taxicab companies.⁴ Each enterprise that pays value-added tax and sales tax amounting to at least RMB 50,000 can apply to win a license plate each year, and can increase its license plates with every additional RMB 500,000 of taxes paid. However, such an entity cannot purchase more than eight vehicles per year. Government-sponsored institutions, social organizations, and other organizations also enter the lottery for these plates.

The license plate lottery is conducted by six applicant representatives, each of whom is a member of either the Beijing Municipal People's Congress, or the Beijing Municipal Committee

of the Chinese People's Political Consultative Conference. Each lottery participant draws a number to generate a six-digit random seed. Under the supervision of notaries and the staff of the Discipline Inspection and Supervision Commission, a technician inputs the lottery numbers sealed by the notary office into a computer, which produces license plate numbers on the spot. The technician records all of the chosen numbers on a disc, copies of which are provided to the notary office, the Beijing Municipal Bureau of Supervision, and the Beijing Municipal Car Quota Control and Administration Office. The chosen numbers are published on the license plate lottery system's website.

Each lottery winner receives a free quota in the form of a downloadable certificate that functions as a permit, allowing the lottery winner the right to purchase a new or used car. Each quota holder may purchase a license plate costing RMB 125. Quotas are nontransferable and cannot be sold. Each winner has six months to purchase a car before their right to buy a car expires. If a lottery winner does not purchase a car during this interval, that permit serves to expand the pool of quotas available in the next lottery.

Those who already own a car do not have to win the lottery; instead, they retain their car registrations. If an existing car owner destroys her car or sells it to a used car dealer, for example, she retains her license plate and can purchase a new car.

Enforcement of the license plate lottery system is strict. To prevent people from purchasing cars in other cities and driving in Beijing, the city issued driving restrictions forcing all cars with outside license plates to obtain official certificates before entering the 5th ring road. Moreover, only cars with Beijing plates are permitted to drive in the area within the 5th ring road during the weekday rush hours of 7:00 a.m. to 9:00 a.m. and 5:00 p.m. to 8:00 p.m. Owners of cars with outside plates found driving during rush hour are fined RMB 100 for each violation.

4. Effects of the lottery system

4.1. Impacts on the demand for license plates

Residents reacted to Beijing's mid-December announcement of the license plate restriction by moving forward vehicle purchase decisions. In the two weeks between the initial announcement and the complete freeze on new vehicle registrations, new vehicle registrations rose more than 50 percent (Fig. 3). News articles at the time described customers at dealerships who purchased vehicles without bargaining and without having the opportunity to select their cars.

Over time, the lottery pool saw dramatic increases in the number of entries and commensurate decreases in the probability of success (Fig. 4). While those who participated in the first lottery in January 2011 had a 9 percent chance of winning, that probability decreased to less than 2 percent by August 2012.

The lottery's setup is likely to have artificially swelled the number of entries. Qualified applicants can enter the lottery at no cost by filling out a short online application. Because the application process is simple and free, it is common for all members of a family to enter the lottery in the hopes of winning a single vehicle. In addition, households without an immediate need for a car often entered the lottery, anticipating that they would need a car by the time they won it. As a result, many of those who won the lottery did not purchase a vehicle within the allotted six months, invalidating their quotas. In June 2012, 10.9 percent of individual lottery winners did not purchase vehicles within the specified period. Even among corporate entrants, 22.8 percent of winners failed to purchase a vehicle.

The difficulty of winning the lottery suggests that many entrants with a high willingness to pay for a vehicle were unable to buy a car. At the same time, many quotas expired without being used.

⁴ In sum, with individuals competing for 17,600 plates, businesses competing for 2000 plates, and commercial car companies competing for 400 plates, 20,000 new vehicle license plates are available each month.

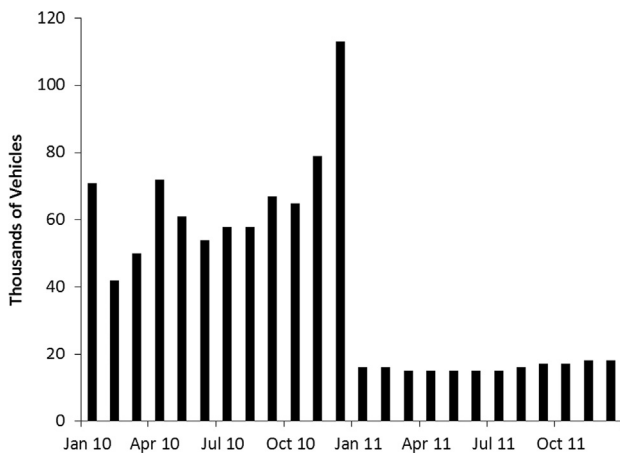


Fig. 3. Monthly changes in numbers of new vehicle registrations in Beijing, 2010 and 2011.

Source: Beijing Municipal Commission of Transport and Beijing Transportation Research Center (2012).

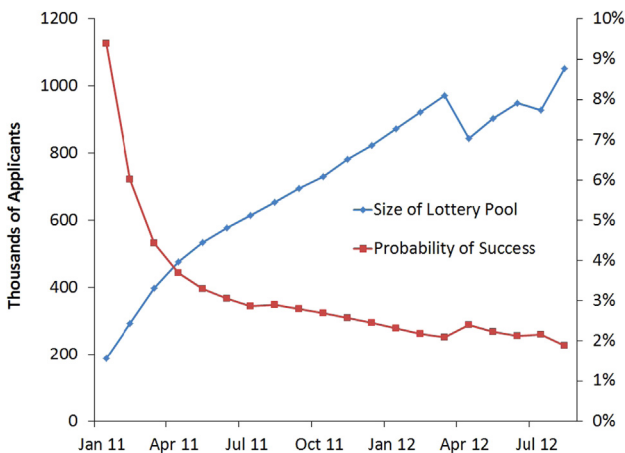


Fig. 4. The growing number of license plate lottery applicants, and the diminishing chance of success.

Source: Beijing Municipal Commission of Transport and Beijing Transportation Research Center (2012).

The high rate of quota expirations is fully rational under Beijing's lottery system. An applicant who wins the lottery has the option to purchase a vehicle, with no financial cost⁵ if the quota expires. If a lottery winner's quota expires, he can simply enter the lottery again at no additional cost. The system is set up so that the act of entering the lottery generates positive value for the entrant in the form of a chance to win an option to purchase a car.

Beijing revised its policy in January 2012 to address this problem. The validity period for applications is now limited: every three months, those participating in the license plate lottery are required to use the lottery website to confirm their interest in obtaining a vehicle; otherwise their applications will be removed. Although this change imposed a relatively low bar to remaining in the lottery, it resulted in thousands of expired applications

⁵ There is no monetary cost to entering the lottery. However, there may be other costs involved, such as the cost of time, and psychological anxiety. The non-monetary cost is individual to the lottery entrant, and depends on the value of the car to the entrant. The anxiety cost might be high for some entrants, such as those who need a car and cannot rely on public transportation. One of our authors participated in the lottery program. Like many others, she entered the lottery at a time when she did not have an immediate need for a car, but hoped that she might have a need for one by the time she won.

Table 1

Total numbers of motor vehicles in Beijing, in Millions.
Source: Beijing Transport Annual Report (2003–2011).

Year	Motor vehicles	Private vehicles	Private cars
2003	2.124	1.389	0.929
2004	2.296	1.547	1.107
2005	2.583	1.798	1.343
2006	2.876	2.065	1.612
2007	3.128	2.358	1.928
2008	3.504	2.693	2.289
2009	4.019	3.186	2.818
2010	4.809	3.909	3.566
2011	4.984	4.028	3.717

Note: Private cars are sedans owned by individuals. Private vehicles are vehicles owned by individuals, including not only sedans, but also coaches, motorcycles and three-wheeled automobiles. Motor vehicles include passenger vehicles and freight vehicles, owned by individuals, companies and government agencies.

beginning in April 2012; we see dramatic fluctuations in the number of active lottery applications starting at this time.

A vigorous black market sprang up in response to the shortage of available license plates. In particular, an illegal trade called "license plate renting" emerged. In this trade, an individual needing a car purchases a vehicle in the name of a lottery winner, paying the winner for this service. Once the individual who originally needed a car obtains his own quota by winning the lottery himself, he transfers the vehicle purchased from the original lottery winner to himself. A second related practice involves car dealerships. Before the freeze on vehicle registrations, some dealerships had hundreds of second-hand cars on hand. They immediately registered for license plates for these vehicles, allowing them to be driven legally after the restriction on vehicle license plates was implemented. An individual needing a car could then purchase a vehicle with plates already attached to it, paying additional fees ranging between 10,000 and 20,000 RMB per year for the privilege of using the dealer's license plates. These individuals were only able to select from the stock of cars owned by the dealer. These arrangements were made with the understanding that the purchaser would return the rented license plate back to the dealer when the purchaser had won the lottery herself.

Consequently, the People's Government of Beijing Municipality sternly declared that no special quotas or quota trading were permitted, with quotas to be recalled in serious cases. Those caught breaking the law would be held legally liable. It is believed that these new regulations are generally obeyed; media reports of black market license plates have diminished in recent months.

4.2. Effects on the quantity of motor vehicles

The license plate lottery dramatically curbed increases in motor vehicle registrations (Fig. 3). New car registrations were cut by over 75 percent between 2010 and 2011. Similarly, the growth in the total number of registered cars was dramatically slowed (Table 1).

Total sales of new vehicles in Beijing dropped commensurately, from 920,000 in 2010 to 400,000 in 2011, a decrease of 57 percent. Retail sales in Beijing decreased from RMB 155.7 billion to RMB 86.8 billion, a decrease of 44 percent. The second-hand automobile market also suffered from the new policy: sales of second-hand automobiles dropped from 520,000 in 2010 to 400,000 in 2011.⁶

Although the motor vehicle market clearly suffered from the policy, the overall economy of the Beijing municipality did not

⁶ Because used cars can be sold in adjacent provinces, second-hand car sales did not drop as much as sales of new cars.

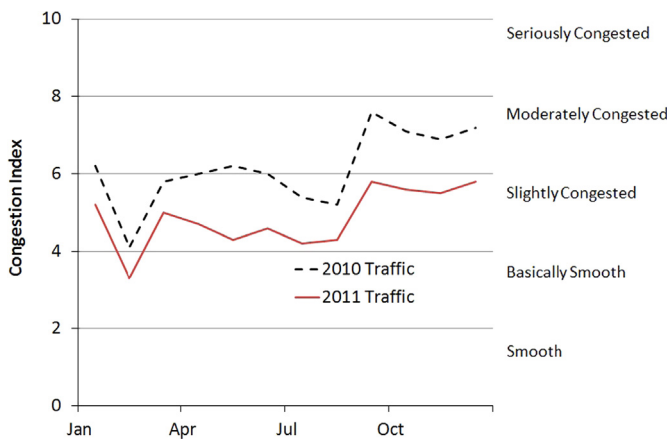


Fig. 5. Congestion index levels before and after the advent of the Beijing License Plate Lottery.

Source: Beijing Municipal Commission of Transport and Beijing Transportation Research Center (2012).

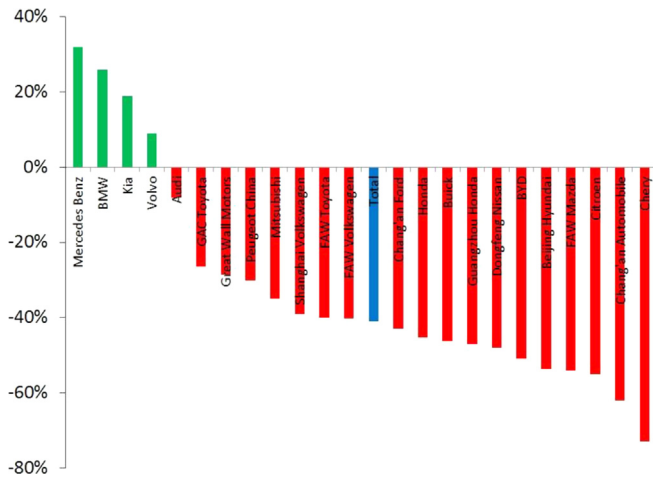


Fig. 6. Changes in sales of cars between 2010 and 2011 in Beijing, by brand of car. Source: China Association of Automobile Manufacturers.

seem to suffer ill effects. Beijing's GDP and GDP per capita continued rising steadily in 2012 without any signs of a slowdown (Fig. 1).

4.3. Improvements in congestion

The advent of the license plate lottery was accompanied by immediate improvements in traffic. Fig. 5 compares congestion in 2010 and 2011. For each month in 2011, congestion is noticeably lower, suggesting that the set of policies enacted during this time had a significant effect. According to the 2011 *Beijing Transport Annual Report*, average driving speeds rose from 22.6 km/h in 2010 to 25.3 km/h in 2011.

Although the advent of the lottery and the improvement in congestion happened at the same time, we cannot attribute the improvement in congestion solely to the lottery. License plate restrictions cannot decrease the number of vehicles on the road, but merely slow the growth of these vehicles (Table 1). Along with the lottery policy, Beijing continued to reinforce its policies targeted at controlling congestion, including upgrades to road capacity, increases to parking fees, and the construction of more bus lanes. These measures had been ongoing in the years before the Beijing lottery, but congestion had continued to swell (Fig. 2).

In sum, license plate restrictions had a direct effect on the growth in automobiles, which may have allowed these other measures to take effect. License plate restrictions reduced the growth in demand for roads, allowing the growth in the supply of roads and transportation alternatives to finally reduce congestion. We are unable to isolate the impact of license plate restrictions on congestion in this paper.

It is too early to judge whether reduced congestion will continue into the medium and long term. The post-Olympics policy restricting the weekdays on which each car could be driven was effective in the short run but lost its effectiveness after about two years. While part of this loss of effectiveness was undoubtedly due to the increased number of cars, part of it had to be attributable to driver adaptations to the car restrictions.

4.4. Changes to the composition of motor vehicles purchased

Statistics from the Beijing Municipal Office of State Administration of Taxation show that the average unit price of automobiles grew 28 percent between 2010 and 2011. Between 2003 and 2010, the average unit price of automobiles increased by a combined 14 percent.

Many possible factors may be responsible for this increase in unit sales price, such as increased incomes and increased numbers of very wealthy residents in Beijing. One additional reason is the lottery policy. Because families have to win the lottery before purchasing a new car, they tend to concentrate their transportation investment into a single high-end car rather than two low-end cars.

Evidence on the distribution of sales by brand supports this interpretation (Fig. 6). Total sales of vehicles declined sharply and, whereas many mid-market firms took punishing double-digit drops in sales, the steepest drops were suffered by domestic Chinese firms, which have more limited brand reputations. Luxury imported brands, like Mercedes-Benz and BMW, actually expanded their sales.

Prior evidence supports the idea that car restrictions cause purchasers to buy more expensive cars. Xiao et al. (2013) analyzed the impact of vehicle quota auction system in Shanghai. They found that conditional on purchase of a license in a quota system, consumers intend to purchase more expensive high-end vehicles.

5. The effect of the lottery on fuel use

In this section of our paper, we provide simple calculations which show how Beijing's lottery policy may affect vehicle sales and the total consumption of fuel.

The total amount of fuel consumed is calculated using this equation:

$$\text{Fuel used} = \text{number of vehicles} \times \text{vehicle distance traveled} \times \text{average fuel consumption rate} \quad (1)$$

Under this framework, we explore three mechanisms by which the lottery policy may impact the amount of fuel consumed. First, it will have a sure impact on the number of vehicles in Beijing. Second, it may have an impact on vehicle distance traveled. Third, it may change the average fuel consumption rate of Beijing's fleet of cars.

In this section, we calculate each of the items on the right-hand side of Eq. (1) between the years 2011 and 2020. In each subsection that follows, we provide guesses on these items under both the lottery policy and a no-policy scenario (NPS).

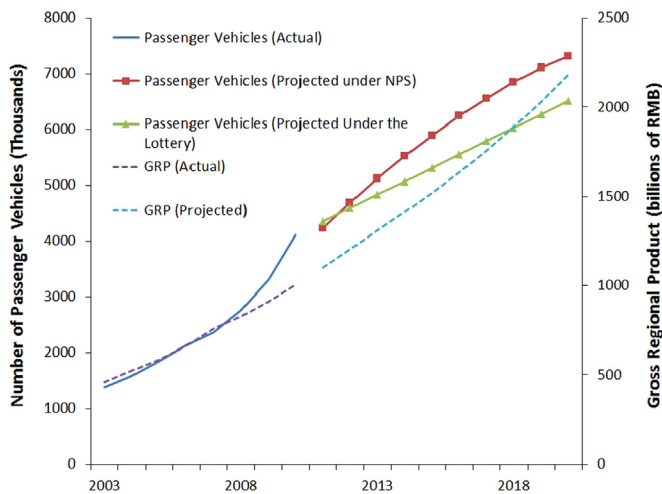


Fig. 7. Projections of the number of passenger vehicles in Beijing.

5.1. Effects of the policy on vehicle growth

The lottery restricted new license plates in Beijing to only 240,000 per year; contrast this with the net increase of 803,000 vehicles in 2010. We plot the actual number of vehicles in Beijing and the number expected under the lottery policy in Fig. 7.⁷

To calculate the counterfactual sales of cars in Beijing under the NPS, we rely on the elasticity of demand for cars with respect to Beijing's GRP. Fig. 7 offers visual evidence that the demand for cars and Beijing's GRP grew in tandem until 2010, when the announcement of the lottery prompted a surge in car sales (Fig. 3).

First, we project Beijing's GRP. Real GRP growth averaged 11.7 percent between 2001 and 2010, falling no lower than 9.1 percent in any year. Beijing's government has predicted that its economy will expand 7.5 percent between 2015 and 2020.⁸ We assume that GRP growth between 2011 and 2015 will fall linearly until it reaches 7.5 percent in 2015.

Next, we calculate the elasticity of growth in the number of vehicles with respect to GRP growth between 2003 and 2010, the years for which we have vehicle data. This elasticity averaged 1.12, with a low of 0.52. We remove 2010 from our calculation, which involved a surge in demand that would not have occurred had the lottery not been enacted. We assume that the elasticity of car demand will be 1.12 in 2010, and that it will fall linearly until it reaches 0.4 under our NPS.

Our calculations are diagrammed in Fig. 7.⁹ The gap between counterfactual growth in vehicle sales and projected growth under the lottery policy is immediately visible in 2011 and remains large and significant over the next 10 years.

5.2. Effects on fuel use

5.2.1. Effects on vehicle distance traveled

We now calculate the second term in Eq. (1), the average vehicle distance traveled (VDT). A few prior studies have also examined this problem in China. He et al. (2005) estimated that

⁷ Note that car owners can replace their existing cars without having to win the lottery. Hence, the annual addition of 240,000 cars under the lottery policy represents a net increase in vehicles.

⁸ Xinhua reported the Beijing Municipal Government's plans on its website at: http://news.xinhuanet.com/fortune/2013-09/26/c_125448188.htm. "The adjustment draft of development goals in Beijing's twelfth five-year plan."

⁹ The astute reader will notice that, in Fig. 9, the number of car sales in our NPS scenario is lower than the number under the lottery policy in 2011. This is because of the surge in demand for cars in 2010, which resulted from the enactment of the lottery.

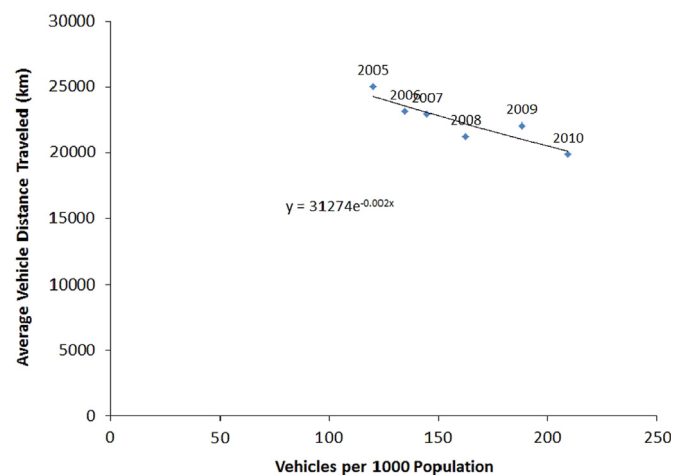


Fig. 8. The relationship between vehicle density and average distance traveled in Beijing.

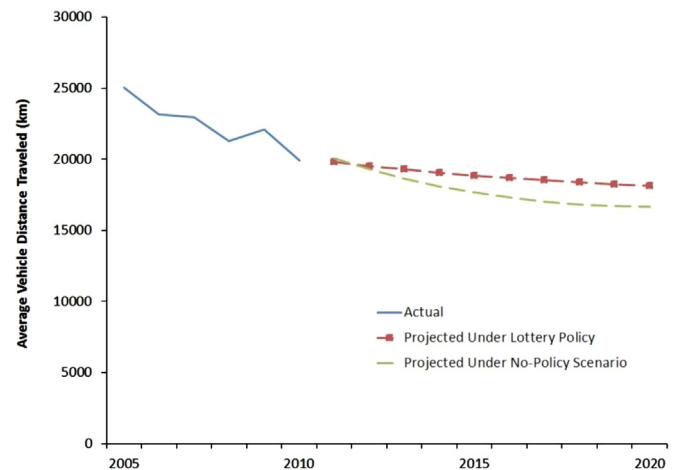


Fig. 9. Projections of vehicle distance traveled in Beijing.

the overall annual VDT of cars decreased only slightly between 1997 and 2002, from 27,200 km to 26,000 km. A more useful analysis for us is that of Hao et al. (2011), who used data from Beijing and projected a much sharper drop based on the aging of the car fleet and shifts in vehicle composition.

We use part of the analysis of Hao et al. (2011) to project VDT in Beijing. Using data from 2003 to 2008, they note that declines in VDT have linearly tracked rates of private car ownership. We analyzed 2005 to 2010 VDT data from the Beijing transport annual report and confirmed this relationship (Fig. 8).

Our strategy is to assume that this linear relationship will continue to hold. We then use this relationship to project VDT based on simple calculations of vehicle ownership rates.

We first calculate private car ownership rates. We assume that Beijing's population will continue to grow at its 10-year average of 2.8 percent per year. Then we use vehicle growth calculations from Section 5.1 to calculate private car ownership rates under both the lottery policy and the NPS. Using these projected vehicle density rates and the linear relationship calculated above, we derive rough calculations of VDT; results are presented in Fig. 9.

Under the NPS scenario, we expect VDT to continue, for the next five years, its recent trend of sharp decline. Gradually, the VDT should flatten out. Under the lottery policy, vehicle density rates do not grow as quickly as they would under the NPS. As a result, VDT continues its decrease, but at a much slower rate.

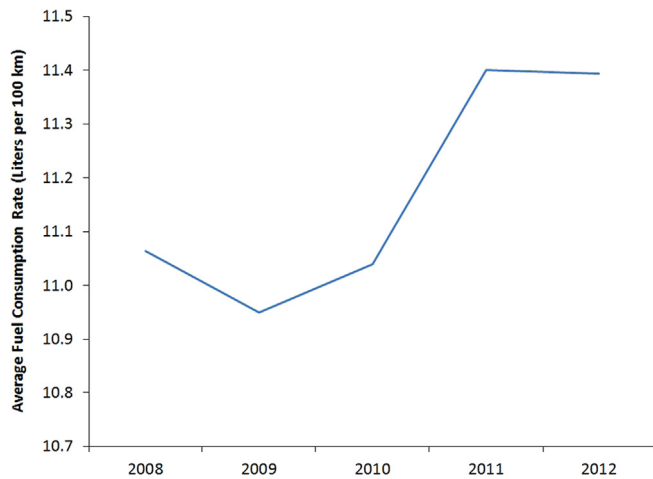


Fig. 10. Calculation of overall average fuel consumption rate in Beijing.

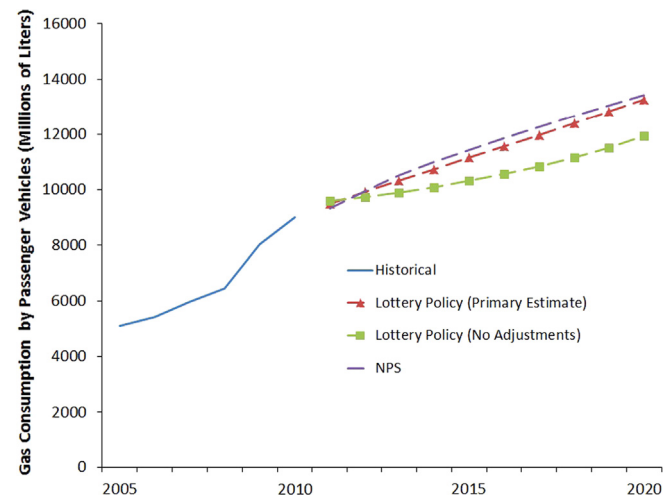


Fig. 11. Calculation of overall fuel consumption in Beijing.

Several reasons may shed light on our projection that the lottery policy will result in a slower fall in VDT. First, congestion decreases the VDT; car owners drive less when driving is less efficient. Because the congestion will increase more slowly under the lottery, falls in VDT will be less steep. A related reason that rising congestion decreases VDT is that substitution in the form of transportation may occur. People may walk, bicycle, or use public transportation more as vehicular congestion rises. Finally, lower overall vehicle ownership rates will result in fewer cars per household; as a result, those cars may be driven more intensively. The lottery limits the ability of households to expand from one car to two cars since household members who already hold car registrations cannot enter the lottery.

5.2.2. Effects on the average fuel consumption rate

As documented in Section 4.4, one effect of the lottery policy was a shift in the composition of motor vehicles purchased. Beijing car buyers moved upmarket, purchasing imported cars and luxury cars at a much higher frequency. Because these cars tend to be larger and heavier, they also tend to be less fuel efficient.

To illustrate this point, Mercedes-Benz, Volvo, BMW, Toyota, and Audi are five of the six least fuel-efficient brands of vehicles. After bouncing up and down between 2008 and 2010, their collective market share grew by 80 percent in 2011 and held steady in 2012. It is likely that the advent of the lottery shifted buyers toward these brands.

Using 2008–2012 data on sales by brand and average fuel consumption rate (AFCR) by brand from the China Association of Automobile Manufacturers, we calculate the AFCR of each year's mix of cars sold in Fig. 10. Based on the shift in sales toward cars with higher AFCR, we calculate that the AFCR of cars sold in Beijing increased by 3.3 percent between 2010 and 2011.

We assume that the AFCR under the NPS would remain at 11 L/100 km, the AFCR we calculated for sales between 2008 and 2010. To calculate the AFCR under the lottery policy, we assume that Beijing will add 400,000 new cars with a higher AFCR mix to its car fleet each year.¹⁰ Using our calculations of the total number of passenger vehicles from Section 5.1, we then calculate the fraction of this fleet that will be composed of the older mix of cars, which we assume to have an AFCR of 11 L/100 km traveled, and the fraction that will be composed of the newer mix of cars, with an AFCR of 11.4 L/100 km traveled. As a result, we calculate that the AFCR of the overall car fleet will increase in an approximately

linear fashion from 11 to 11.2 L/100 km between 2010 and 2020 under the lottery policy.

5.2.3. Effects on total fuel use

Using Eq. (1) and our calculations from Sections 5.1, 5.2.1, and 5.2.2, we calculate three scenarios. First, we calculate the fuel used under the NPS. Second, we calculate the fuel used under the lottery policy. We also calculate, as a third scenario, the lottery policy without other adjustments. In this scenario, we calculate the fuel used when the number of vehicles has been restricted by the lottery, but use the VDT and the AFCR calculated from the NPS.

The results of these simple calculations are displayed in Fig. 11. Consider first the fuel used under the NPS and under the lottery policy scenario with no adjustments. The wide gap between these two lines reflects the effect of restricting the number of cars sold through the lottery policy. The total fuel consumed is expected to decline in the years immediately following the lottery policy, reflecting the continuation of the sharp downward trend in VDT.

Now consider the fuel used under our primary calculation of the lottery policy. Although the amount of fuel consumed in Beijing by passenger vehicles is lower than it would be with no lottery policy in place, this line is far closer to the NPS than it is to the lottery policy scenario with no adjustments. In 2020, the total number of vehicles under the lottery policy is 11 percent lower than the number under the NPS; however, total fuel consumed is only 1 percent lower.

Two factors are responsible for this disparity. First, we anticipate that VDT will fall less sharply under the lottery policy than it would under the NPS. Second, the lottery policy has encouraged consumers to concentrate their purchases on more expensive cars, resulting in a gradual shift in the vehicle mix toward less fuel-efficient vehicles. These factors have the combined effect of mitigating the expected fuel savings from cutting the number of vehicles under the lottery policy.

5.3. Robustness checks

In this section, we present a set of robustness checks which modify central assumptions in the above analysis. In each of these checks, we modify one key assumption of the procedure outlined above, and calculate its impact on the calculated number of vehicles and fuel consumption in 2020. Results are presented in Table 2.

In line 1 of this table, we present our central result, that the lottery policy will decrease the number of vehicles on the roads of

¹⁰ Beijing sold 400,000 vehicles in 2011, the first year of the lottery.

Table 2

Robustness checks of fuel used calculations.

Source: Authors calculations.

Assumption in central case	Modified assumption	Change in vehicles in 2020 (%)	Change in fuel use in 2020 (%)
Base case: all central assumptions	No change in assumptions	–11	–1
GRP growth declines linearly to 7.5% by 2015, grows 7.5% between 2015 and 2020	GRP growth stays at 10.3% between 2010 and 2020	–24	–3
GRP growth declines linearly to 7.5% by 2015, grows 7.5% between 2015 and 2020	GRP growth declines linearly to 3.0% in 2020	–1	0
Elasticity of vehicle growth with respect to GRP growth declines linearly to 0.4 in 2020	Elasticity of vehicle growth with respect to GRP growth declines linearly to 0.8 in 2020	–24	–3
Elasticity of vehicles with respect to GRP growth declines linearly to 0.4 in 2020	Elasticity of vehicles with respect to GRP growth declines linearly to 0.1 in 2020	0	0
AFCR increases linearly from 11 to 11.2 L/100 km in 2020	AFCR increases to 11.4 L/100 km in 2020	–11	0
AFCR increases linearly from 11 to 11.2 L/100 km in 2020	AFCR stays at 11.0 L/100 km until 2020	–11	–3

Note: The change in vehicles represents the calculated decrease in the number of vehicles between the No Policy Scenario in 2020, and the number of vehicles with the lottery policy in place in 2020. The change in fuel use is the decrease or increase between what we calculate for the amount of fuel consumed between the NPS in 2020 and with the lottery in place in 2020.

Beijing by 11 percent in 2020, but will decrease total fuel consumed by only 1 percent. This is the base case for this analysis, and acts as a point of comparison for other scenarios.

In lines 2 and 3, we change the assumptions on GRP growth in Beijing. Rather than assume that GRP growth will decline to 7.5 percent, we assume a high growth case, where GRP growth stays at 10.3 percent, and a low growth case, where growth declines to 3.0 percent in 2020. These growth scenarios change the number of vehicles that would be demanded in the no policy scenario. If GRP growth is high, more vehicles are demanded; if GRP growth is low, fewer vehicles would be demanded. As a result, the number of vehicles under the lottery policy restrictions would be cut 24 percent if GRP growth is high, but only 1 percent is GRP growth is low.

We find that, despite the large change in the number of vehicles on the road, changes in fuel use are comparatively small, a theme running throughout these robustness tables. Increasing the number of vehicles on the road substantially decreases the VDT for each vehicle under the relationship outlined in Section 5.2.1 above. As a result, under the NPS, large increases in the number of cars do not result in large increases in the amount of fuel consumed, and the amount of fuel saved under the lottery policy is comparatively small.

Lines 4 and 5 of Table 2 present our checks on the elasticity of vehicle growth with respect to GRP growth. When this elasticity is high, the number of vehicles in the NPS increases quickly, and the lottery policy has a comparatively greater impact in the number of vehicles on the road.

Similarly to our results on GRP growth, changing the elasticity of vehicle growth with respect to GRP growth has large effects on the number of vehicles that we project in 2020. Again, these increases in the number of vehicles are offset by decreases in VDT for each vehicle, resulting in comparatively minor effects on the amount of fuel demanded.

Lines 6 and 7 of Table 2 present our checks on the AFCR. In our high scenario, we examine the consequences of our calculation if AFCR rises to 11.4 in 2020; in our low scenario, we examine the consequences if AFCR remains at 11.0 in 2020. Under our simple calculations, changes in AFCR have no impact on the number of vehicles demanded, but do change the amount of fuel used. Fuel use is cut 1 percent in the central scenario; it remains unchanged and is cut 3 percent in the high AFCR and low AFCR scenarios, respectively.

The general takeaway from these robustness checks is that assumptions which affect changes in the number of vehicles, such

as GRP growth and vehicle growth, are largely offset by changes in VDT when calculating the amount of fuel that will be consumed. The driving force behind our finding that the Beijing lottery will have smaller effects on fuel consumed than on the number of vehicles in Beijing is this declining relationship between the number of vehicles and the VDT for each vehicle.

6. Discussion and conclusion

We analyze in this paper a unique policy: the use of a lottery to allocate vehicle license plates in Beijing. To conclude, we highlight three key findings from this study.

First, the lottery policy has had a significant effect on the number of vehicles sold. Beijing vehicle sales dropped sharply and the growth in the number of registered vehicles on the road slowed immediately. In combination with other policies, congestion and average driving speeds in Beijing improved at the same time that the lottery was introduced.

Second, the lottery policy might not decrease fuel consumption as much as one might expect. Our projections suggest that, whereas the total number of vehicles in Beijing will decrease 11 percent by 2020, the annual consumption of fuel will decrease only 1 percent. This disparity occurs because the lottery blunts expected increases in vehicle density rates, causing VDT to fall less rapidly than expected.

Finally, the lottery has had the unintended consequence of allocating vehicles to people who clearly do not have the highest willingness to pay. The number of entries in Beijing rapidly dwarfed the number of license plates available until the chance of winning dropped to only 2 percent by August 2012. Since each entrant has the same chance to win the lottery, even a resident with very high willingness to pay may not be able to purchase a vehicle. At the same time, a large number of lottery winners fail to buy vehicles. Winning the lottery is similar to receiving the option to purchase a car; this has positive value since the financial cost of entering the lottery is zero.

Acknowledgement

We are grateful to Sida (through the Environment for Development Initiative at Gothenburg University) for financial support for this research.

References

- Beijing Transportation Research Center, 2002–2013. Beijing Transport Annual Report. Beijing, China.
- Beijing Municipal Commission of Transport and Beijing Transportation Research Center, 2012. Work Report of Thoughts and Measures on Controlling the Total Quantity of Motor Vehicles in the Face of International Cities. Beijing, China.
- Chen, X., Zhao, J., 2013. Bidding to drive: car license auction policy in Shanghai and its public acceptance. *Transp. Policy* 27, 39–52.
- Davis, L.W., 2008. The effect of driving restrictions on air quality in Mexico City. *J. Polit. Econ.* 116, 38–81.
- Deng, H.Y., Ma, A., 2010. Market structure and pricing strategy of China's automobile industry. *J. Ind. Econ.* 58 (4), 818–845.
- Eskeland, G.S., Feyzioglu, T., 1997. Rationing can backfire: the 'day without a car' in Mexico City. *World Bank Econ. Rev.* 11 (3), 383–408.
- Hao, H., Wang, H., Ouyang, M., 2011. Comparison policies on vehicle ownership and use between Beijing and Shanghai and their impacts on fuel consumption by passenger vehicles. *Energy Policy* 39, 1016–1021.
- He, K.B., Huo, H., Zhang, Q., He, D.Q., An, F., Wang, M., Walsh, M.P., 2005. Oil consumption and CO₂ emissions in China's road transport: current status, future trends, and policy implications. *Energy Policy* 33, 1499–1507.
- IBM., 2010. The Globalization of Traffic Congestion: IBM 2010 Commuter Pain Survey. IBM, Armonk, NY.
- Li, Shanjun, 2014. Better Lucky Than Rich? Welfare Analysis of Automobile License Allocations in Beijing and Shanghai. Available at SSRN: (<http://ssrn.com/abstract=2349865>) or 10.2139/ssrn.2349865.
- Liu, D.M., Gao, S.P., An, X.H., 2008. Distribution and source apportionment of polycyclic aromatic hydrocarbons from atmospheric particulate matter PM_{2.5} in Beijing. *Adv. Atmos. Sci.* 25, 297–305.
- Manville, M., Shoup, D., 2005. Parking, people, and cities. *J. Urban Plann. Dev.* 131 (4), 233–245.
- The Central People's Government of the People's Republic of China, 2001. The Interim Regulations of People's Republic of China on Vehicle Purchase Taxes. Accessed online on March 18, 2014 at (http://www.gov.cn/banshi/2005-08/19/content_24868.htm).
- Xiao, J., Zhou, X., Hu, W., 2013. Vehicle Quota System and its Impact on the Chinese Auto Markets: A Tale of Two Cities. Available at SSRN: (<http://ssrn.com/abstract=2294217>).
- Xie, X., Zhang, S., Yi, R., Wu, D., Huang, D., 2011. The social cost of traffic congestion in Beijing" (in Chinese). *China Popul. Resour. Environ.* 21 (1), 28–32.